

1/14 tqec

stimdx demo (Tom Ginsberg): support for dynamic circuits

Builds treelike data structures, separable from the compiler

Use lambda functions as conditionals

Coin flip operator

stim blocks are static, relies on Clifford gates

Conversion between frameworks

How are we going to use this

Michael Perlin — QLDBC

Sliding window decoding features for TQEC

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link to the demo (

<https://colab.research.google.com/drive/1CwOSL9Z0xFccORqLk7SAm2BqkNUAK4Zc?usp=sharing>

) also on the github

<https://github.com/tomginsberg/stimdx/tree/main>

Slide deck: Compiling an algo. to the surface code

Current block graph doesn't implement node color

Take a block graph, convert current notation into colors. Low code ways to specify that

Simplify eg XXZ

Tianyi/Adrien

Further: specify entire structures of graph, not just the node color.

Patch rotation (Gilad)

Patch rotation block (sketchup) that can be converted into circuits. good beginner tasks to work on

Updated surface code paper inbound

Kabir: walking codes, fast sliding circuit design

Issue 491

Updates: slow sliding circuits

Bella: stretch stabilizers (#657), visualization

J/Tianyi: subgraph prefabrication, exponential scaling, streaming

Subgraphs with relative nodes and can be joined or repeated (eg, adder)

Specify repeating sections

In general: Preemptively make an issue for something you want to work on so everyone knows.

Alexandru: feedback on PR

Ioana: Pandora: streamlining compilation

Topologiq: Space contained compilation vs unconstrained

Beams: a way to implement ZX graphs in 3d

Problem: need to pack a graph in a box

Change beams to arbitrary mathematical objects

Want to create bidirectional beams

Issues for new people: get in touch with people who have experience with the project to get involved.

Right now we focus on 2d qubit arrays with nearest neighbor interactions (for superconducting qubits). With ion traps/neutral atom architectures, we can handle other interactions (between any pair of qubits). Create an issue for this

Omar: Survey of existing compilers